

NOOR_MOHAMMAD

Cairo, Egypt



Professional Summary

Computer Science at CHI with strong foundations in Python, algorithms, object-oriented programming, and software engineering, pursuing a specialization in Artificial Intelligence and Machine Learning. Experienced in developing structured technical projects using Python, machine learning workflows, and data analysis techniques. Passionate about AI systems, predictive analytics, and backend engineering, with a strong focus on problem solving, debugging, and building scalable intelligent applications.

Education

CHI

Bachelor of Computer Science

Jul-2023 to Jul-2027

Egypt

Relevant Coursework

- Data Structures
- Algorithms
- Artificial Intelligence
- Object-Oriented Programming
- Databases
- Operating Systems
- Computer Networks
- Software Engineering

Projects

F1 Race Strategy Optimization System | *Python, Machine Learning, Data Analysis*

2025 – Present

- Developing an AI-driven Formula 1 race strategy system that analyzes historical race telemetry and tire performance data to predict optimal pit-stop timing and tire selection strategies.
- Designing and implementing a scalable data pipeline in Python for collecting, preprocessing, and structuring real-world Formula 1 racing datasets from multiple sources.
- Building machine learning models to evaluate race strategy outcomes based on tire degradation, lap-time trends, weather conditions, and race variables.
- Applying predictive analytics and feature engineering techniques to simulate race decision scenarios and improve strategic performance modeling.
- Utilizing Python, data engineering workflows, and analytical visualization tools to support experimentation, debugging, and ML model evaluation.

Advanced Sudoku Solver | *Python, Algorithms, OOP*

Planned Personal Project

- Designing a modular Sudoku solving engine using recursive backtracking and optimization techniques to solve puzzles efficiently.
- Implementing performance tracking features including execution time analysis, backtracking statistics, and puzzle difficulty evaluation.
- Applying object-oriented programming principles and clean software architecture practices to improve maintainability and scalability.
- Documenting project structure, algorithms, and implementation details using GitHub and technical README documentation.

Technical Skills

Programming Languages: Python, JavaScript, Java, SQL

Machine Learning & Data: Machine Learning, Data Analysis, Pandas, NumPy, Predictive Modeling, Data Preprocessing

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook

Concepts: Algorithms, Data Structures, Object-Oriented Programming, Software Engineering, Problem Solving

Currently Learning: scikit-learn, Linux, FastAPI, Data Pipelines

Certifications & Learning

Google Machine Learning Crash Course | *Google Developers*

In Progress

Kaggle Machine Learning Learning Path | *Kaggle*

In Progress

Activities & Interests

- AI Engineering, Machine Learning Systems, Problem Solving